1. Initial exploration:

Here we'll first try to explore the data, understand it and answer some questions with simple queries

1.a. Data type of columns in a table

```
-- Checking datatypes of tables
SELECT column_name, data_type
FROM `targetsql-373323.target_sql.INFORMATION_SCHEMA.COLUMNS`
WHERE table_name = 'customers';
```

Row	column_name	data_type
1	customer_id	STRING
2	customer_unique_id	STRING
3	customer_zip_code_prefix	INT64
4	customer_city	STRING
5	customer_state	STRING

* INFORMATION_SCHEMA.COLUMNS is a special function in bigquery

1.b. Get the time period for which the data is given

```
-- Time range between which the orders were placed
SELECT min(order_purchase_timestamp) AS first_order,
  max(order_purchase_timestamp) AS last_order
FROM `targetsql-373323.target_sql.orders`;
```

Row	first_order	last_order
1	2016-09-04 21:15:19 UTC	2018-10-17 17:30:18 UTC

1.c. Number of cities and states in our dataset

```
-- Count the Cities & States of customers who ordered during the given period.
SELECT count(distinct(geolocation_city)) as city_count,
```

```
  count(distinct(geolocation_state)) as state_count
FROM `target_sql.geolocation`;
```

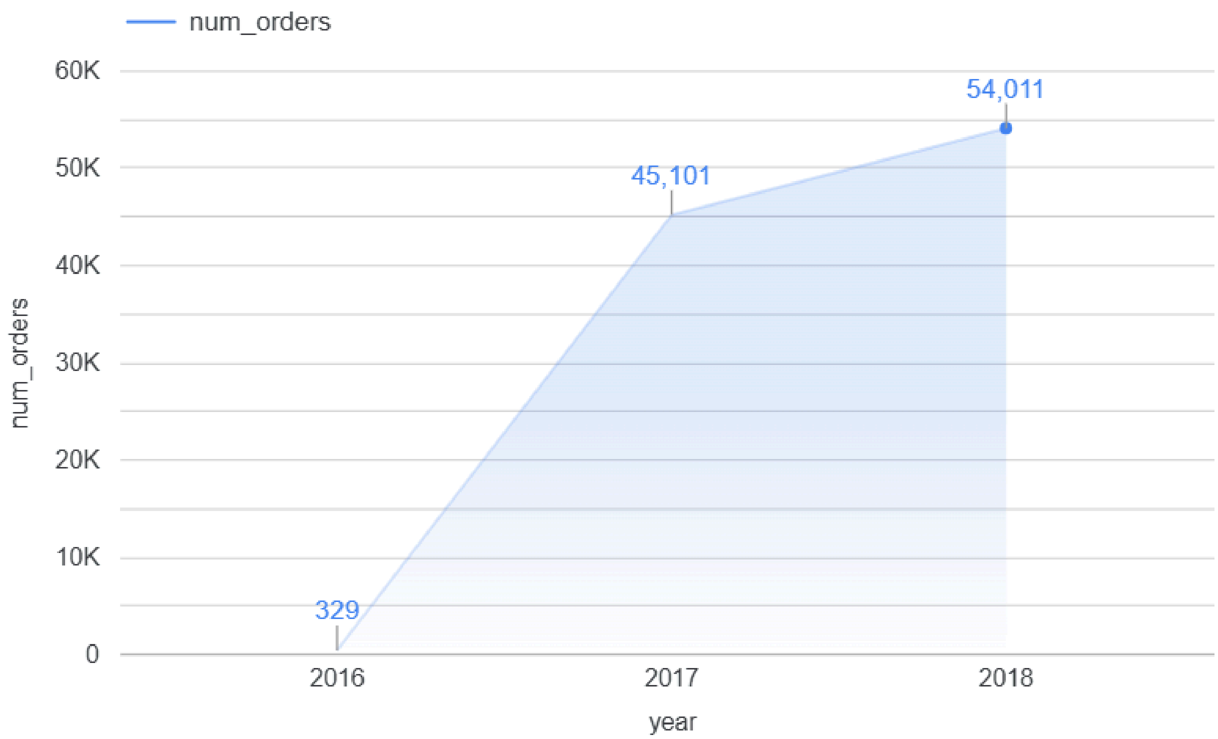
Row	city_count	state_count
1	8011	27

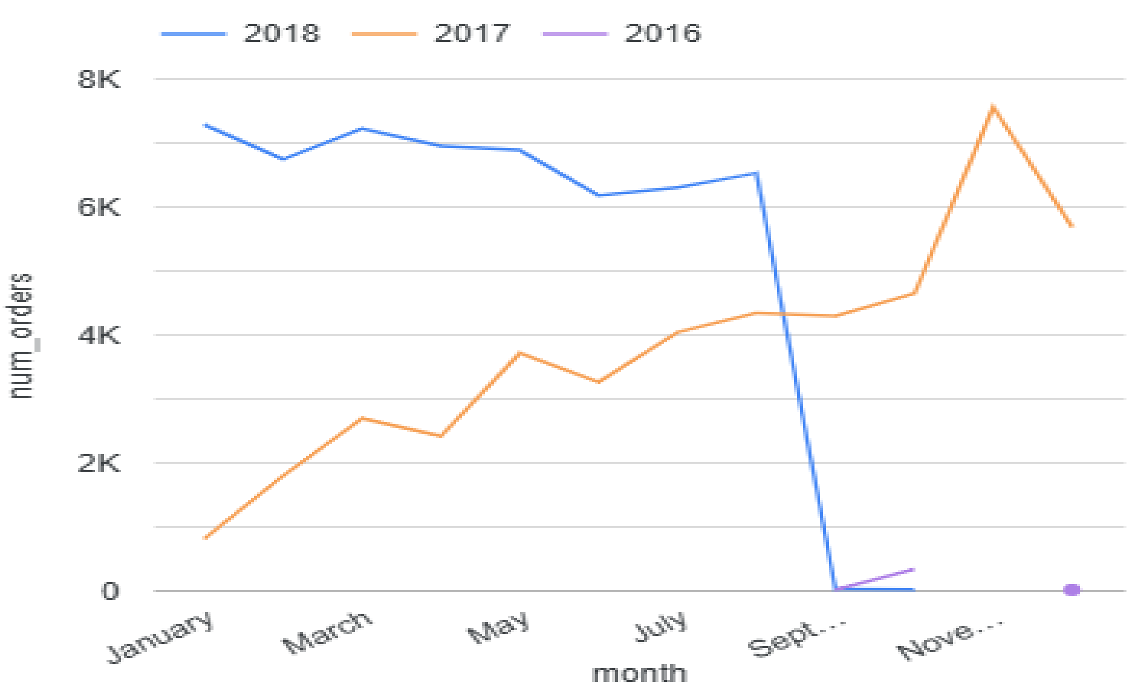
2. In-depth Exploration:

2.a. Is there a growing trend in e-commerce in Brazil? How can we describe a complete scenario?

```
--Trend in the no. of orders placed over the past years.
SELECT EXTRACT(year from order_purchase_timestamp) AS year,
  EXTRACT(month from order_purchase_timestamp) AS month,
  count(*) AS num_orders
FROM `target_sql.orders`
GROUP BY year,month
ORDER BY year,month;
```

Row	year	month	num_orders
1	2016	9	4
2	2016	10	324
3	2016	12	1
4	2017	1	800
5	2017	2	1780
6	2017	3	2682
7	2017	4	2404
8	2017	5	3700



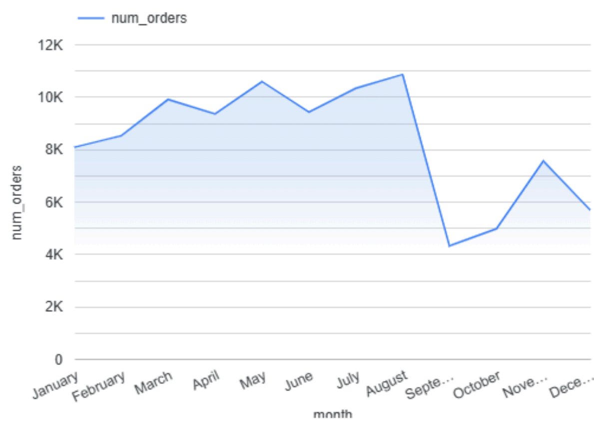


Continuous rise in orders month over month. In general we can see clearly that customers are more prone to buy things online than before.However the increase in 2017 was higher. 2018 orders are actually falling and need further analysis.

2b. Can we see some seasonality with peaks at specific months?

```sql
--Trend in the no. of orders placed over the months.
SELECT EXTRACT(month from order_purchase_timestamp) AS month,
  count(*) AS num_orders
FROM `target_sql.orders`
GROUP BY month
```

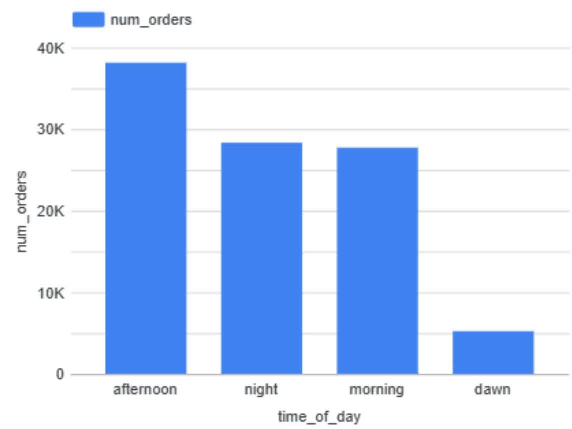
	month ▲	num_orders
1.	January	8,069
2.	February	8,508
3.	March	9,893
4.	April	9,343
5.	May	10,573
6.	June	9,412
7.	July	10,318
8.	August	10,843
9.	September	4,305
10.	October	4,959
11.	November	7,544
12.	December	5,674



2c.During what time of the day, do the Brazilian customers mostly place their orders? (Dawn, Morning, Afternoon or Night)?

```sql
--Trend by the hour of the day (Dawn, Morning, Afternoon, Night)
SELECT
  CASE WHEN EXTRACT(hour FROM order_purchase_timestamp) BETWEEN 0 and 6 THEN 'dawn'
    WHEN EXTRACT(hour FROM order_purchase_timestamp) BETWEEN 7 and 12 THEN 'morning'
    WHEN EXTRACT(hour FROM order_purchase_timestamp) BETWEEN 13 and 18 THEN
'afternoon'
    ELSE 'night'
  END AS time_of_day,
  Count(*) as num_orders
  FROM `target_sql.orders`
  GROUP BY time_of_day
  ORDER BY num_orders DESC;
```

	time_of_day	num_orders
1.	afternoon	38,135
2.	night	28,331
3.	morning	27,733
4.	dawn	5,242



3. Evolution of E-commerce orders in Brazil region

Now we'll try to understand data based on state or city level and see what variations are present and how the people in various states order and receive deliveries.

3.a. Get month on month orders by states.

```
--Get month on month orders by states.

SELECT c.customer_state,
  EXTRACT(month from
order_purchase_timestamp) as month,
  COUNT(*) AS num_orders
FROM `target_sql.orders` o
JOIN `target_sql.customers` c
on o.customer_id=c.customer_id
GROUP BY c.customer_state, month
ORDER BY num_orders DESC;
```

Row	customer_state	month	num_orders
1	SP	8	4982
2	SP	5	4632
3	SP	7	4381
4	SP	6	4104
5	SP	3	4047
6	SP	4	3967
7	SP	2	3357
8	SP	1	3351
9	SP	11	3012

Different states have different monthly total orders, it is important to analyse each state separately.

3.b. Distribution of customers across the states in Brazil

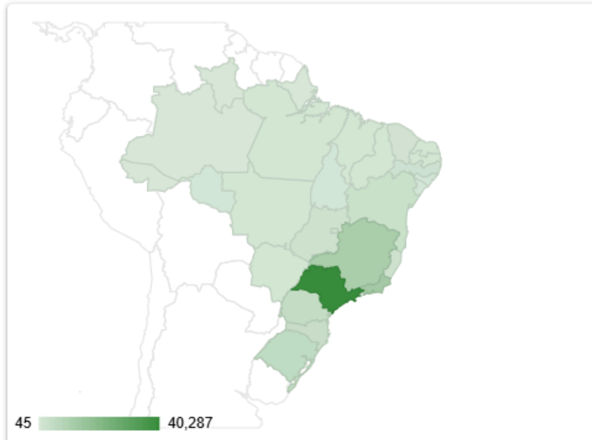
```
--Distribution of customers across the states in Brazil
SELECT g.geolocation_state,
  COUNT(DISTINCT(c.customer_unique_id)) AS num_customers
FROM `target_sql.customers` c
JOIN `target_sql.geolocation` g
```

```sql
ON c.customer_zip_code_prefix=g.geolocation_zip_code_prefix
GROUP BY g.geolocation_state
ORDER  BY num_customers DESC;
```

	geolocation_state	num_customers ▾
1.	SP	40,287
2.	RJ	12,372
3.	MG	11,248
4.	RS	5,284
5.	PR	4,871
6.	SC	3,547
7.	BA	3,268
8.	ES	1,959
9.	GO	1,944
10.	DF	1,913
11.	PE	1,605

1 - 27 / 27

geolocation_state by num_customers



Majority of customers are located in Sao Paulo, Rio De Janeiro and Minas Gerais while lowest number of customers in Roraima, Amapá, Acre.

4. Impact on Economy:

Analysis of the money movement by e-commerce by looking at order prices, freight and others.

4.a. % increase from 2017 to 2018

```sql
--% Increase in cost from 2017 to 2018
WITH base AS(
  SELECT EXTRACT(year from o.order_purchase_timestamp) AS year,
    EXTRACT(month from o.order_purchase_timestamp) AS month,
    SUM(p.payment_value) as cost,
  FROM `target_sql.orders` o
  JOIN `target_sql.payments` p
  ON o.order_id=p.order_id
  WHERE EXTRACT(year from o.order_purchase_timestamp) BETWEEN 2017 AND 2018
    AND EXTRACT(month from o.order_purchase_timestamp) BETWEEN 1 AND 8
  GROUP BY year,month
  ORDER BY year,month
),
base_2 AS(
  SELECT year,
  SUM(cost) as cost
  FROM base
```

```
  GROUP BY year
  ORDER BY year
),
base_3 AS(
SELECT * , LAG(cost,1) OVER(ORDER BY year) as last_year_cost
FROM base_2
)
SELECT *,
  ROUND((cost-coalesce(last_year_cost,0))/coalesce(last_year_cost,1) *100,2) || '%' as
per_inc
FROM base_3
WHERE year= 2018;
```

Row	year	cost	last_year_cost	per_inc
1	2018	8694733.840000...	3669022.119999...	136.98%

There is a 137% increase in cost of orders from 2017 to 2018 (for Jan-Aug months).

4.b. Sum and mean price and freight by customer state

```
--Total & Average value of order price,order freight for each state
WITH CTE AS
  (SELECT oi.order_id,oi.price, oi.freight_value, o.customer_id, c.customer_state
  FROM `target_sql.order_items` oi
  JOIN `target_sql.orders` o
  ON oi.order_id=o.order_id
  JOIN `target_sql.customers` c
  ON o.customer_id = c.customer_id)

SELECT customer_state,
sum(price) as sum_price,sum(price)/count(distinct(order_id)) as mean_price,
sum(freight_value) as sum_freight_value ,sum(freight_value)/count(distinct(order_id))
as mean_freight_value,
FROM CTE
GROUP BY customer_state;
```

Row	customer_state	sum_price	mean_price	sum_freight_value	mean_freight_value
1	SP	5202955.049999...	125.7511794561...	718723.0700000...	17.37095033232...
2	RJ	1824092.669999...	142.9315679360...	305589.3100000...	23.94525231155...
3	PR	683083.760000001	136.6714205682...	117851.6799999...	23.57976790716...
4	RS	750304.0200000...	138.1266605301...	135522.7399999...	24.94895802650...
5	MG	1585308.030000...	137.3274454261...	270853.4600000...	23.46270443520...
6	GO	294591.9499999...	146.7822371699...	53114.98000000...	26.46486297957...
7	SC	520553.3399999...	144.1177574750...	89660.25999999...	24.82288482834...
8	PB	115268.0800000...	216.6693233082...	25719.73	48.34535714285...

5. Analysis on sales, freight and delivery time

5.a Days between purchasing, delivering and estimated delivery

```
--Days between purchasing,estimated,actual delivery date
select order_id,
```

```sql
TIMESTAMP_DIFF(order_delivered_customer_date,order_purchase_timestamp, DAY) AS
time_to_delivery,
TIMESTAMP_DIFF(order_delivered_customer_date,order_estimated_delivery_date, DAY) AS
diff_estimated_delivery
from `target_sql.orders`
where order_status='delivered';
```

Row	order_id	time_to_delivery	diff_estimated_d...
1	bfbd0f9bdef84302105ad712db...	54	36
2	98974b076b01553d49ee64679...	43	-6
3	c4b41c36dd589e901f6879f25a...	36	-14
4	d2292ff2201e74c5db154d1b7a...	29	-20
5	95e01270fcbae986342340010...	30	-19

5.b Top 5 states with highest/lowest average time to delivery

```sql
--Top 5 states with highest/lowest average time to delivery
select g.geolocation_state as state,
SUM(TIMESTAMP_DIFF( order_delivered_customer_date,order_purchase_timestamp,
DAY))/COUNT(ORDER_ID) AS avg_dil_time,
from `target_sql.orders` o
inner join `target_sql.customers` c
on o.customer_id=c.customer_id
inner join `target_sql.geolocation` g
on c.customer_zip_code_prefix=g.geolocation_zip_code_prefix
where order_status='delivered'
group by state
order by avg_dil_time
limit 5
```

Row	state	avg_dil_time
1	SP	8.468892914352...
2	PR	11.03876404770...
3	MG	11.41821678372...
4	DF	12.49651789233...
5	SC	14.48408434580...

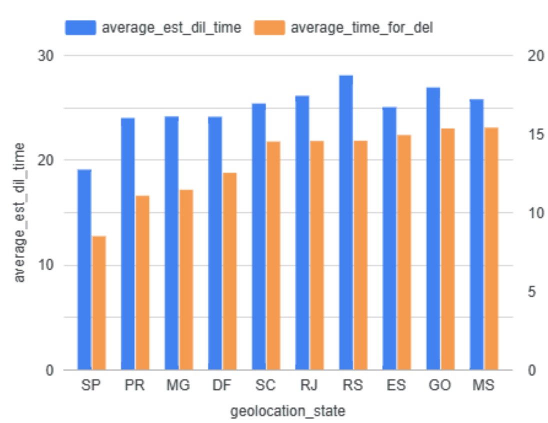
São Paulo has lowest mean time to delivery while Roraima has highest mean time to delivery.

5.c. Top 5 states where delivery is really fast/slow compared to estimated date

```
--Top 5 states where delivery is really fast/ not so fast compared to estimated date
select geolocation_state,
SUM(TIMESTAMP_DIFF(order_delivered_customer_date,order_purchase_timestamp,
DAY))/COUNT(ORDER_ID) AS average_time_for_del,
SUM(TIMESTAMP_DIFF(order_estimated_delivery_date, order_purchase_timestamp,
DAY))/COUNT(ORDER_ID) AS average_est_dil_time,
from `target_sql.orders` o
inner join `target_sql.customers` c
on o.customer_id=c.customer_id
inner join `target_sql.geolocation` g
on c.customer_zip_code_prefix=g.geolocation_zip_code_prefix
where order_status='delivered'
group by geolocation_state
order by (average_time_for_del-average_est_dil_time)
```

	geolocation...	average_est_d... ▲	average_time_fo...
1.	SP	19.05	8.47
2.	PR	23.97	11.04
3.	DF	24.08	12.5
4.	MG	24.11	11.42
5.	ES	25.01	14.89
6.	SC	25.35	14.48
7.	MS	25.75	15.37
8.	RJ	26.08	14.52
9.	GO	26.88	15.31
10.	TO	27.9	16.3
11.	RS	28.04	14.53

1 - 27 / 27



5.d. Average time for delivery

```
--Avg time for delivery
WITH cte as
  (select
order_id,order_purchase_timestamp,order_delivered_customer_date,order_estimated_delive
ry_date,
  DATE_DIFF(order_delivered_customer_date,order_purchase_timestamp, DAY) AS
time_to_delivery,
  DATE_DIFF(order_delivered_customer_date,order_estimated_delivery_date, DAY) AS
diff_estimated_delivery,
  FROM `target_sql.orders`)
select avg(time_to_delivery) as avg_time_to_delivery,
avg(diff_estimated_delivery) as avg_diff_estimated_delivery
from cte;
```

Row	avg_time_to_deliv...	avg_diff_estimate...
1	12.09408557568...	-10.9580102823...

6. Payment type analysis

6.a. Count of orders for different payment types Month over Month

```
--Orders for different payment types Month over Month
SELECT payment_type, COUNT(o.order_id) as order_count,
Extract( month from order_purchase_timestamp) as month,
Extract( year from order_purchase_timestamp) as year,
FROM `target_sql.payments` p
join `target_sql.orders` o
on o.order_id=p.order_id
GROUP BY payment_type, year, month
order by year, month;
```

Row	payment_type	order_count	month	year
1	credit_card	3	9	2016
2	voucher	23	10	2016
3	debit_card	2	10	2016
4	UPI	63	10	2016
5	credit_card	254	10	2016
6	credit_card	1	12	2016
7	UPI	197	1	2017
8	debit_card	9	1	2017

6.b. Count of orders based on the no. of payment installments

```
--Count of orders based on the no. of payment installments
SELECT distinct(payment_installments) as installments, count(order_id) as Num_orders,
FROM `target_sql.payments`
where payment_installments>1
GROUP BY payment_installments
order by Num_orders desc;
```

Row	installments	Num_orders
1	2	12413
2	3	10461
3	4	7098
4	10	5328
5	5	5239
6	8	4268
7	6	3920
8	7	1626
9	9	644

6.c.What percentage of orders were canceled or unavailable

```
--What percentage of orders were canceled or unavailable
select counter/total*100 as canc_per from (
```

```sql
select sum(case when order_status in ('canceled','unavailable') then 1 else 0 end) as
counter,
count (1) as total from `target_sql.orders`
);
```

Row	canc_per
1	1.240936836918...

6.d. Find customers with more than one order

```sql
--Find customers with more than one order
SELECT customer_unique_id, count(1) as ordercount FROM
`target_sql.customers` c
JOIN `target_sql.orders` o ON c.customer_id = o.customer_id
GROUP BY customer_unique_id
HAVING COUNT(order_purchase_timestamp) >1
ORDER BY COUNT(order_purchase_timestamp) DESC ;
```

Row	customer_unique_id	ordercount
1	8d50f5eadf50201ccdcedfb9e2ac8455	17
2	3e43e6105506432c953e165fb2acf44c	9
3	ca77025e7201e3b30c44b472ff346268	7
4	6469f99c1f9dfae7733b25662e7f1782	7
5	1b6c7548a2a1f9037c1fd3ddfed95f33	7
6	47c1a3033b8b77b3ab6e109eb4d5fdf3	6
7	f0e310a6839dce9de1638e0fe5ab282a	6
8	12f5d6e1cbf93dafd9dcc19095df0b3d	6
9	dc813062e0fc23409cd255f7f53c7074	6
10	[illegible]	6

Insights -
1) Sales are highest in November
2) Sales are highest from 12-6pm
3) Majority of customers are located in Sao Paulo, Rio De Janeiro and Minas Gerais while lowest number of customers in Roraima, Amapá, Acre
4) The average time to delivery (between purchase and delivery) is about 12 days.

5) The average time between estimated and actual delivery is about 10 days i.e. products arrive 10 days earlier than expected on an average.
6) Roraima has highest mean of freight value.
7) São Paulo has lowest mean of freight value.
8) Roraima has highest mean time to delivery.
9) São Paulo has lowest mean time to delivery.
10) Maximum number of orders were placed with just a single payment installment.
11) There is a 137% increase in cost of orders from 2017 to 2018 (for Jan-Aug months).

Recommendations-
1) Stock up inventory in November
2) Clearance sale in November
3) Offer more deals during afternoon peak hours
4) More deals, varieties and offers in states having majority customers like Sao Paulo, Rio De Janeiro and Minas Gerais
5) More deals and promotions in states like Sao Paulo where time to delivery is lowest.